

Saurabh Gupta

Data Scientist

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Summary

Results-oriented Data Scientist with 8+ years of experience in leveraging advanced analytics, machine learning, and deep learning to extract actionable insights from large, complex datasets. Proven success in solving business challenges by designing and deploying scalable, end-to-end data-driven solutions. Skilled in Python, Pytorch, SQL and cloud platforms, with a track record of delivering measurable improvements in operational efficiency and revenue growth. Experienced in collaborating with cross-functional teams and mentoring junior analysts, driving innovation through cutting-edge machine learning models. Committed to continuous learning and staying ahead of industry advancements.

Professional Experience

Phenom pvt ltd - Bangalore

Product Development Engineer II

Jan 2025 - Present

- Edge has been acquired by Phenom, integrating Edge's capabilities into the Phenom platform.

AVR Edge Networks Pvt Ltd. - Bangalore

Feb 2018 - Dec 2024

- Data Scientist - II (Dec 2020 - Dec 2024)**
 - Led a team of three data scientists, overseeing project delivery, mentoring team members, and ensuring adherence to best practices in machine learning and data engineering.
 - Designed and developed a **Skill Intelligence Assistant** leveraging **OpenAI**, **LangGraph**, and **Chainlit** to enable users to efficiently manage and structure skill taxonomy.
 - Developed multiple **OpenAI LLM-based prompts** to automate and optimize key processes, including job description generation and data validation, resulting in improved system efficiency and enhanced user experience.
 - Fine-tuned the **LLaMA 2 model** using **QLoRA** for skill recommendation, achieving a **10%** improvement in recommendation accuracy.
 - Designed and developed a **Knowledge Graph** by modifying the translation model to recommend skill and title ontologies, resulting in a 20% improvement in hit@10 accuracy.
 - Fine-tuned the **Sentence Transformer** model on demand-supply pairs to enhance matching, resulting in a 10% improvement in Mean Average Precision (MAP).
 - Designed and implemented a skill and title normalization engine, enhancing underlying capabilities and improving model performance through **Sentence Transformer** training.
 - Optimized the skill extraction engine by training the **LSTM** based model, resulting in a 5% improvement in supply recommendation accuracy.
- Machine Learning Engineer (Feb 2018 - Nov 2020)**
 - Trained **Word2Vec** and **GloVe** models on job description data to identify related skills and titles, resulting in a **15%** improvement in model performance.
 - Enhanced **demand forecasting engine** to predict future demand for upcoming quarters, enabling clients to optimize supply allocation and improve resource planning.
 - Developed **descriptive and prescriptive dashboards** to provide insights into demand and supply, helping clients identify critical demands and internal fulfilment opportunities, resulting in reduced hiring costs and improved cost efficiency.
 - Implemented a **Streamlit**-based monitoring dashboard to capture system metrics, providing clarity on system performance and measuring the impact of changes.
 - Designed and developed interactive **PowerBI** dashboards, empowering clients to make data-driven decisions and significantly enhancing user experience.

IBM India pvt ltd - Noida

Associate System Engineer

Jan 2017 - Jan 2018

- Designed and implemented an **ETL pipeline** to streamline data extraction, transformation, and loading from multiple sources, reducing data processing time by 20%.
- Performed **exploratory data analysis (EDA)** and data cleaning, applying statistical techniques to derive actionable insights, resulting in **5%** improvement in model performance.

Skills

Programming & Libraries

Python, PyTorch, Hugging Face, Transformers, LLM, PEFT, LangChain, LangGraph, Scikit-learn, Pandas, NumPy

Machine Learning

Data Science, AI, Statistics, Exploratory Data Analysis(EDA), Feature Engineering, Descriptive & Predictive Modeling, NLP, Time-Series Analysis, Recommender Systems

Tools & Deployment

Jupyter, Git, Docker, DVC, MIFlow, CICD, Kubernetes

Data Visualization

PowerBI, Matplotlib, Seaborn, Streamlit

Database Technologies

SQL, MongoDB, Elastic Search, Postgres

Cloud Platforms

AWS, Google Cloud, Azure

Projects

Fine Tuning of LLM models

- Fine-tuned LLaMA 2 and GEMMA models using QLoRA techniques on a sample dataset.
- Enhanced skill-matching tasks, improving model efficiency and performance.

<https://github.com/gupta24789/fine-tuning-llms>

Fine Tuning of Sentence Transformer

- Optimized SentenceTransformer models on task-specific datasets to improve semantic similarity detection.
- Achieved enhanced performance in text retrieval and ranking tasks.

https://github.com/gupta24789/sentence-transformers/blob/main/03_train_st.ipynb

Named Entity Recognition

- Developed and optimized deep learning models (LSTM, CNN + LSTM, SpaCy) for named entity recognition.
- Achieved an 88% F1-score after rigorous training and validation, surpassing baseline performance.

<https://github.com/gupta24789/named-entity-recognition>

Emotions Detection

- Utilized BOW and TFIDF techniques for baseline emotion detection (F1-score: 75%).
- Implemented advanced deep learning models (RNN, LSTM, fine-tuned transformer models), achieving a 91% F1-score.

<https://github.com/gupta24789/sentiment-analysis>

Duplicate Question Pairs

- Developed a Siamese network for duplicate question detection.
- Achieved a 74% F-score for effective duplicate pair identification.

<https://github.com/gupta24789/siamese-networks>

Education

M Tech (NIT Bhopal)

Computer Science (9.03 CGPA)

B Tech (Ideal Institute of technology, Ghaziabad)

Computer Science (73.98%)

Certifications

Deep Learning Specialization - Coursera

This specialization covers neural networks, CNNs, RNNs, sequence models, and deep learning frameworks such as TensorFlow. Developed expertise in model optimization, hyperparameter tuning, and applying deep learning techniques to real-world challenges.

NLP Specialization - Coursera

This specialized focused on natural language processing, covering text processing, language models, sentiment analysis, and sequence labeling. Gained hands-on experience with state-of-the-art NLP techniques and frameworks, applying them to real-world tasks.